MILITARY SPECIFICATION

COMMUNICATIONS SETS, UHF
AN/WSC-3(V) SATELLITE AND LOS VERSIONS

This amendment forms a part of Military Specification MIL-C-28838C(EC), dated 21 October 1985, and is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 3

* 2.1.1, after "MIL-M-38510", add:

"SPACE AND NAVAL WARFARE SYSTEMS COMMAND (SPAWAR)

ELEX-S-425	System (Semiautomatic), UHF Demand Assigned Multiple Access (DAMA)
ELEX-M-619	Mini-DAMA Communications Set, AN/USC-42(V)1, Shipboard Version Full Scale Development Specification".

* 2.1.1, after "MIL-STD-188", add:

"MIL-STD-188-114 Electrical Characteristics Of Digital Interface Circuits".

2.1.2, under PUBLICATIONS, NATIONAL SECURITY AGENCY (NSA): Delete "CSEEB-32"

After "CSESD-14", add:

"ASSISTANT SECRETARY OF NAVY (SUPPLY AND LOGISTICS)

NAVMAT Memorandum Of Nuclear Hardening Criteria
18 April 1985".

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2.1.2: Delete:

"DEFENSE NUCLEAR AGENCY (DNA)

DNA 1420H	TREE (Transient Radiation Effect Of Electronics) Handbook
DNA 2114H	EMP (Electromagnetic Pulse) Handbook".

* 2.1.2: Delete "WS 6536D" and substitute "WS 6536E with SCN 1".

2.1.2: Delete:

"SPACE AND NAVAL WARFARE SYSTEMS COMMAND (SPAWAR)

NAVELEX PME 116 11019/SAL Ser. 00684 of 5 March 1976	EMP and TREE Hardening Specification, Nuclear Environment Statement

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The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
1	Reprinted without change
2	2

AMSC N/A

FSC 5820

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"BUREAU OF NAVAL PERSONNEL

NAVPers 18068 Manual of Qualifications For Advancement".

2.1.2, after "NAVSEA 0967-LP-143-4010", add:

"NAVAL PERSONNEL COMMAND

NAVPERS 18068D Navy Enlisted Manpower And Personnel Classifications And Occupational Standards".

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* 3.1.1, line 3: Delete "OK-327" and substitute "OK-367".

* 3.1.1, after last sentence, add:

"The AN/WSC-3(V)15 version shall be capable of interfacing with the UHF Demand Assigned Multiple Access (DAMA) multiplexers TD-1271B/U (specified in ELEX-S-425) and AN/USC-42(V)1 (specified in ELEX-M-619).".

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3.1.20 through 3.1.20.1: Delete and substitute:

"3.1.20 Nuclear hardening. The survivability of the equipment under nuclear electromagnetic pulse (EMP) and transient radiation effects in electronics (TREE) shall be required in the design and construction of the equipment. The equipment shall survive the nuclear hardness levels for surface combatant ships specified in enclosure (1) to NAVMAT memorandum of 18 April 1985.

* 3.1.20.1, added by Amendment 1, lines 2 and 3. Delete "3.1.20.1.1" and "3.1.20.1.2" and substitute "MIL-STD-461, Requirements CS11 and RS05, as applicable".

* 3.1.20.1.1, added by Amendment 1: Delete.

* 3.1.20.1.2, added by Amendment 1: Delete.

3.1.20.2: Delete and substitute:

"3.1.20.2 TREE. The equipment shall operate under a TREE environment and survive the nuclear hardness levels specified in 3.1.20 without permanent damage or interruption of service beyond the duration of the transient. Degradation due to neutrons or total dose shall not cause the equipment to perform below allowable limits for operation as specified herein.".

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3.2, line 2: Delete "3.2.1.18" and substitute "3.2.18.2".

* After 3.2.4.6, add new paragraph:

"3.2.4.7 Transmit center frequency adjustment. The AN/WSC-3(V)12 and AN/WSC-3(V)15 shall provide a strap-selectable adjustment of the RF component to a value within ±60 Hz, ±30 Hz, or ±0 Hz frequency range about the nominal, selected center frequency when the equipment is operated in nonDAMA and DAMA PSK modes. This discrete frequency adjustment shall occur each time the transmitter is keyed and shall be either a random or pseudo-random process. Minimum frequency adjustment shall not exceed 0.5 Hz. A pseudo-random implementation shall not permit the frequency adjustment to repeat within a continuous equipment operating time of 720 hours in the specified modes. Conformance to the frequency stability requirements of 3.2.4.3 shall be maintained during transmit operation.".

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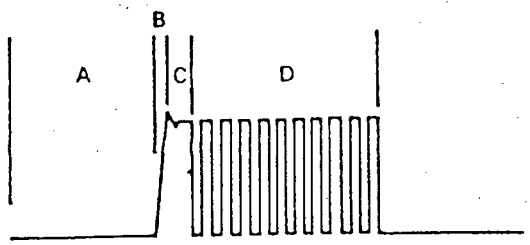
* 3.2.6.1.3: Delete text and substitute:

"The transmitter RF power output to the antenna group from the AN/WSC-3(V)12 and AN/WSC-3(V)15 shall be as specified in 3.2.6.1.3.1 and 3.2.6.1.3.2 operating in the nonDAMA SATCOM 25 kHz mode, or any of the nonDAMA SATCOM OFFSET kHz communication modes. The equipment shall not be damaged in any way if the transmitter is keyed while the equipment is channeling.".

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* After 3.2.6.1.3, add new paragraphs and new FIGURE 0:

"3.2.6.1.3.1 Transmitter key delay time. The transmitter key delay time shown in FIGURE 0 shall be dithered in a random manner not to exceed 40 ms ±10 ms to mask the worst case variation of the transmitter key delay time.



A. Transmitter key delay time
B. Transmitter rise time
C. Transmit data delay time
D. PSK modem preamble

FIGURE 0. Transmitter waveform.

"3.2.6.1.3.2 Transmit data delay time. The transmit data delay time shown in FIGURE 0 shall be dithered in a random manner to mask the worst case variation of the transmit data delay time between equipment."

* 3.2.6.8, line 2: Delete "3.2.6.8.3" and substitute "3.2.6.8.4".

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* After 3.2.6.8.3, add new paragraph:

"3.2.6.8.4 SAS switch. The equipment shall incorporate a switch to permit selection of either SAS or nonSAS secure voice operation.".

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3.2.7.14, line 2: Delete "3.5.7.14.1.1 through 3.5.7.14.3" and substitute "3.2.7.14.1 through 3.2.7.14.3".

3.2.7.14.1, line 1: Delete "3.2.7.14.1" and substitute "3.2.7.14.1.1".

* 3.2.7.14.1.1.2, title and line 1: Delete "output power" and substitute "audio output power".

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* 3.2.7.14.1.2.2, title: Delete "output level" and substitute "audio output level".

* 3.2.7.14.1.2.3, line 1: Delete "output impedance" and substitute "output load impedance".

* 3.2.7.14.2.2.2, title: Delete "frequency response" and substitute "audio output level".

* 3.2.7.14.2.2.2, line 1: Delete "audio output power" and substitute "audio output level".

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* 3.2.7.16.1.2, line 1: Delete "3 dBm" and substitute "-3 dBm"; line 2: Delete "as applied" and substitute "is applied".

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3.2.8.5.4.2, line 2: Delete "CSEEB-32 and".

3.2.8.6, line 1: Delete "3.2.8.6" and substitute "3.2.8.6.1".

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* 3.2.8.7.3, after last sentence, add:

"When the AN/WSC-3(V)12 and AN/WSC-3(V)15 operate in the nonDAMA modes, the PSK modem shall enable baseband inputs for transmission upon completion of the RF power output delay specified in 3.2.6.1.3.".

* After 3.2.8.7.3.4, add new paragraph:

"3.2.8.7.3.5 Preamble. The preamble shown in FIGURE 0 shall be a fixed integral length dependent upon the data rates of 3.2.8.7.3.1.".

* 3.2.8.8, title: Delete "FSK" and substitute "PSK".

* 3.2.8.8, line 1: Delete "FSK" and substitute "PSK".

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* 3.2.17.7, line 2: Delete "specified TABLE III" and substitute "following additional".

* 3.2.18, line 2: Delete "AN/WSC-3(V)12" and substitute "AN/WSC-3(V)3".

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* After 3.2.18.2, add new paragraphs:

"3.2.19 AN/WSC-3(V)15 characteristics. The equipment shall conform to the requirements specified herein for the AN/WSC-3(V)2 and AN/WSC-3(V)3, except as specified in 3.2.19.1 through 3.2.19.7.

"3.2.19.1 Transmitter performance. The transmitter performance of the equipment shall be as specified herein, except as specified in 3.2.19.1.1 through 3.2.19.1.5.

"3.2.19.1.1 RF output power. Upon receipt of a transmit command from the DAMA multiplexer, the RF output power shall be stabilized in amplitude to within ±1 dB in 800 µs maximum. The RF output power shall be less than 100 milliwatts until the frequency has settled to within ±5 kHz of the final steady-state frequency.

"3.2.19.1.2 RF frequency settling time. Upon receipt of a transmit command from the DAMA multiplexer, the transmitter RF output frequency shall settle to within ±50 Hz of the final steady-state frequency within 800 µs maximum.

"3.2.19.1.3 Noise power output. With the transmitter keyed, but with no 70-MHz IF applied to the 70-MHz input of the equipment, the noise power density at the RF output of the radio shall be not greater than -55 dBm per Hz. There shall be no spurious frequency output greater than -55 dBm in a frequency range of ±12.5 kHz with respect to the center frequency.".

"3.2.19.1.4 DAMA keyline. The DAMA keyline shall be TTL compatible with a Low or ground for transmit and a High or open for receive. A Low shall be 0.8 VDC maximum and a High shall be 2.4 VDC minimum.

"3.2.19.1.5 Keyline voltage. Keyline voltage shall be supplied by the equipment to the DAMA multiplexer on the keyline in the high or receive state. The keyline voltage supplied by the equipment shall be +2.0 VDC to +7.0 VDC and limited to a maximum current of 25 mA.

"3.2.19.2 Receiver performance. The receiver performance of the equipment shall be as specified herein except as specified in 3.2.19.2.1 through 3.2.19.2.4.

"3.2.19.2.1 IF output bandwidth. With the IF centered at 70 MHz, the IF output amplitude, over ±25 kHz from 70 MHz center frequency, shall be within 1 dB of the center frequency value.

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"3.2.19.2.2 Internal 70 MHz crosstalk. The internal attenuation between the 70 MHz input and the 70 MHz output shall be 77 dB or greater.

"3.2.19.2.3 RF input and 70 MHz output. The 70 MHz output shall be stabilized in amplitude to within ±1 dB and in frequency to within ±50 Hz of the final value within 800 μs after the RF frequency is applied to the receiver front end. The 70 MHz output shall decay to the noise level within 1 μs after termination of the RF input.

"3.2.19.2.4 70 MHz output level. With the radio operated in the DAMA mode and a signal generator having a source impedance of 50 ohms and an open circuit RF output of 3.5 μV (-96.0 dBm) connected to the receiver antenna input connector through a 50-ohm transmission line, the 70 MHz output when measured into a 50-ohm termination shall be greater than 5.6 μV (-92.0 dBm).

"3.2.19.3 DAMA or nonDAMA strap. An internal strap which selects either the DAMA or nonDAMA mode of operation shall be provided in the equipment chassis.

"3.2.19.4 BITE. DAMA-modified equipment shall have no internal BITE capabilities when operated in either the DAMA or nonDAMA mode for the following positions:

Position	Test
13	BLANKER MODULE
14	MAIN IF
16	FM DETECTOR
18	DATA IF
20	FSK DETECTOR
21	PSK-LOGIC-RECEIVER
22	FSK DETECTOR
23	DATA BUFFER HI LEVEL
24	DATA BUFFER LO LEVEL

BITE may be tested in these positions by connecting a 40 dB attenuator between the 70 MHz input and 70 MHz output connectors.

"3.2.19.5 Remote tuning requirements for DAMA and nonDAMA. When operated in the DAMA mode, the input to the equipment's five preset tuning lines shall be low level unbalanced in accordance with MIL-STD-188-114. When operated in the nonDAMA mode, the input to the equipment's five preset tuning lines shall be TTL compatible. Input signal sensing shall be as specified in a through c in the DAMA mode:

a. A positive input to the equipment with respect to ground shall be a binary 1 state.

b. A negative input to the equipment with respect to ground shall be a binary 0 state.

c. The input voltage level shall be between 4 VDC to 6 VDC.

"3.2.19.6 Switchable offsets. Position 5 of the internal offset switch on the equipment shall be configured for zero offset selection.

"3.2.19.7 Remote interlock. When operating in the DAMA or nonDAMA mode, the equipment shall provide a remote interlock circuit which shall inhibit equipment remote control specified in 3.2.11 and 3.2.19.5, unless the circuit is grounded. An open circuit shall inhibit remote control of the equipment."

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3.7.5.2, line 3: Delete "NAVPERS 130689" and substitute "NAVPERS 18068D".

* 3.8.2: Delete and substitute:

"3.8.2 Workmanship screen. All equipment shall withstand a defect detection vibration screen of random type vibration at $0.04g^2/Hz$ ±3 dB from 80 Hz to 350 Hz and temperature cycling as specified in 4.8.1.2.

"3.9 Rescreening of semiconductor, microelectronic and hybrid devices. All semiconductor, microelectronic, and hybrid devices shall be rescreened at -55°C, +25°C, and +125°C. The applicable device specification parameters for each device shall be measured at each temperature.".

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4.3, last line: Delete "TABLE V" and substitute "TABLE VII".

* TABLE VII, "Secure voice interface (transmit)" entry, under "Requirement paragraph" column: Delete "3.2.6.8 through 3.2.6.8.3" and substitute "3.2.6.8 through 3.2.6.8.4".

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* TABLE VII, "Digital data" entry, under "Requirement paragraph" column: Delete "3.2.8.7 through 3.2.8.7.3.4" and substitute "3.2.8.7 through 3.2.8.7.3.5".

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* TABLE VII, after "Alternate intake and remote BITE" entry: Add the following new entry:

			First article inspection	Group B
"AN/WSC-3(V)15 characteristics	3.2.19 through 3.2.19.7	4.5.1	X	X".

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4.5.5: Delete and substitute:

"4.5.5 Verification of nuclear hardening. Verification of compliance of the requirements for the nuclear environment is divided into two categories: a) nuclear EMP, and b) nuclear ionizing radiations. Each requires a separate set of analyses and tests as specified in 4.5.5.1 and 4.5.5.2.".

* 4.5.5.1, added by Amendment 1: Delete and substitute:

"4.5.5.1 EMP. The EMP testing shall be as specified in MIL-STD-462, Test Methods CS11 and RS05.".

* 4.5.5.1.1 through 4.5.5.1.2.4, added by Amendment 1: Delete.

Add 4.5.5.2:

"4.5.5.2 Nuclear radiation effects testing. Verification of conformance of hardness to nuclear radiation effects shall be demonstrated either by analyses or nuclear simulation tests. Electronic piece-parts and circuits for which nuclear radiation test data are available on the specific device can be verified by analyses. For devices in the same family, verification shall be by analysis at 10 times the specified levels or by testing. For devices with no applicable test data, verification shall be by testing. Test plans submitted to the Government shall include the following: facility to be used; parts, circuits, or equipments to be irradiated; levels of irradiation; diagnostic techniques; dosimetry and calibration; and description of data reduction methodology. For direct injection tests, test plans shall include test waveform; interface circuits to be tested; test equipments; test fixtures; diagnostic techniques; and description of data reduction.".

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* 4.8.1.2.2, line 6: Delete "reaches 80 percent" and substitute "come within 10°C".

* FIGURE 9: Delete "+75" in two places and substitute "+71".

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* 6.2.2, delete the change made by Amendment 1 and substitute new 6.2.2:

"6.2.2 Data requirements. When this specification is used in and acquisition and data are required to be delivered, the data requirements identified below should be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List, incorporated into the contract. When the provisions of DoD Federal Acquisition Regulations Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below should be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph No.	Data requirement title	Applicable DID No.
3.4	Request For Confirmation Of Nomenclature	DI-CMAN-80194
3.4	Request For Assignment Of Serial Number And Serial Number Prefix Letter	DI-CMAN-80195
3.4	Request For Approval Of Identification Plate Drawings	DI-CMAN-80196
3.5	Safety Assessment Report	DI-SAFT-80102
3.6	Reliability Program Plan	DI-R-7079
3.7	Maintainability Progam Plan	DI-R-7103
3.7.5.2	Recommendations, Maintenance Level	UDI-L-22340
4.1.2	Quality Program Plan (QPP)	DI-R-5297A
4.3	First Article Inspection Report	DI-T-4902
4.4	Equipment Test Plan (NONSYSTEM)	DI-T-3709A
4.5.1	Procedures, Test	UDI-T-22710B
4.6	Reliability Test Procedures	DI-RELI-80251
4.7.1	Plan, Maintainability Demonstration	DI-R-2129
4.7.1	Report, Maintainability Demonstration	DI-R-7113".

* FIGURES 10, 11, and 12, added by Amendment 1: Delete.

Preparing activity:
NAVY - EC

(Project No. 5820-N855(EC))

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